

Vectors

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22nd September 2025

Vectors

Vectors are geometric objects that have magnitude and direction

We will use (x, y) to denote a point in $\mathbb{R}^2$ and $\langle x, y \rangle$ to denote a vector in $\mathbb{R}^2$

- Geometrically a vector is a directed line segment
- Algebraically it is determined by its coordinates

$\mathbb{R}^n$

$$\begin{aligned}\mathbb{R}^n &= \{\vec{v} = \langle v_1, v_2, \dots, v_n \rangle \mid v_1, v_2, \dots, v_n \in \mathbb{R}, n \in \mathbb{N}\} \\ &= \text{set of all ordered } n\text{-tuples of real numbers}\end{aligned}$$

Unit Vectors

A unit vector is a vector of length 1 for a given vector $\vec{v} \in \mathbb{R}^n$. The direction of $\vec{v}$ is $\frac{1}{\|\vec{v}\|}$

Basis Vectors

$\{\langle 1, 0 \rangle, \langle 0, 1 \rangle\}$ are the standard basis vectors in $\mathbb{R}^2$

$\{\langle 1, 0, 0 \rangle, \langle 0, 1, 0 \rangle, \langle 0, 0, 1 \rangle\}$ are the standard basis vectors in $\mathbb{R}^3$

These are often denoted by $\vec{i}, \vec{j}, \vec{k}$ respectively

Operations Defined on Vectors

Magnitude of a vector.

The magnitude of a vector $\vec{v} = \langle v_1, v_2, \dots, v_n \rangle \in \mathbb{R}^n$ is defined to be the distance from $(v_1, v_2, \dots, v_n)$ to the origin in $\mathbb{R}^n$

$$\|\vec{v}\| = |\vec{v}| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

Addition of vectors

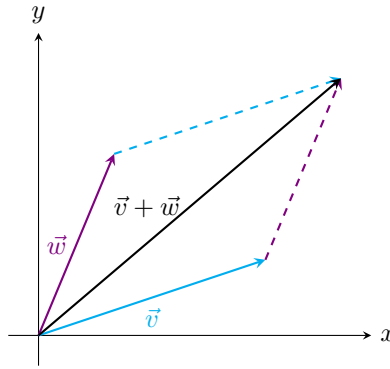
Suppose $\vec{v}, \vec{w} \in \mathbb{R}^n$

$$\vec{v} = \langle v_1, v_2, \dots, v_n \rangle$$

$$\vec{w} = \langle w_1, w_2, \dots, w_n \rangle$$

We define $\vec{v} + \vec{w} = \langle v_1 + w_1, v_2 + w_2, \dots, v_n + w_n \rangle$

- Algebraically we add components
- Geometrically $\vec{v} + \vec{w}$ is the directed line segment that is a diagonal of the parallelogram formed by $\vec{v}$ and $\vec{w}$



Scalar multiplication of vectors

A scalar is a real number, $\lambda \in \mathbb{R}$

Let $\lambda \in \mathbb{R}, \vec{v} \in \mathbb{R}^n$,

$\lambda \vec{v}$ is defined to be

$$\langle \lambda v_1, \lambda v_2, \dots, \lambda v_n \rangle$$

- Geometrically $\lambda \vec{v}$ points in the same direction as $\vec{v}$ if $\lambda > 0$ and in the opposite direction if $\lambda < 0$ and $\lambda \vec{v}$ has magnitude $|\lambda| \|\vec{v}\|$

Dot Product (Scalar Product)

Let $\vec{v}, \vec{w} \in \mathbb{R}^n$

$$\vec{v} = \langle v_1, v_2, \dots, v_n \rangle$$

$$\vec{w} = \langle w_1, w_2, \dots, w_n \rangle$$

The dot product of $\vec{v}$ and $\vec{w}$ is denoted by $\vec{v} \cdot \vec{w}$ and is defined as follows

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2 + \dots + v_n w_n \in \mathbb{R}$$

Geometrically the dot product can be defined as

$$\vec{v} \cdot \vec{w} = \|\vec{v}\| \|\vec{w}\| \cos \theta$$

where θ is the angle ($\leq \pi$) between the two vectors

Orthogonality

Two vectors $\vec{v}, \vec{w} \in \mathbb{R}^n$ are said to be orthogonal if and only if $\vec{v} \cdot \vec{w} = 0$, i.e. the angle between the two vectors is $\frac{\pi}{2}$

Cross Product (Vector Product)

Note that the cross product is only defined in $\mathbb{R}^3$

Properties of the operations

Let $\vec{v}, \vec{w}, \vec{q} \in \mathbb{R}^n, \lambda \in \mathbb{R}$

- $\vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{v}$
- $\lambda(\vec{v} + \vec{w}) = \lambda \vec{v} + \lambda \vec{w}$
- $\lambda(\vec{v} \cdot \vec{w}) = (\lambda \vec{v}) \cdot \vec{w} = \vec{v} \cdot (\lambda \vec{w})$
- $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2$
- $\vec{v} \cdot \vec{w} = \|\vec{v}\| \|\vec{w}\| \cos \theta$
- $\vec{v} \cdot (\vec{w} + \vec{q}) = \vec{v} \cdot \vec{w} + \vec{v} \cdot \vec{q}$

Proofs of the Properties

1.

$$\begin{aligned}\vec{v} \cdot \vec{w} &= v_1 w_1 + v_2 w_2 + \cdots + v_n w_n \\ &= w_1 v_1 + w_2 v_2 + \cdots + w_n v_n \quad (\text{Commutativity of multiplication of real numbers}) \\ &= \vec{w} \cdot \vec{v}\end{aligned}$$

2.

$$\begin{aligned}\lambda(\vec{v} + \vec{w}) &= \lambda(v_1 + w_1 + v_2 + w_2 + \cdots + v_n + w_n) \\ &= \lambda(\vec{v}_1 + \vec{v}_2 + \cdots + \vec{v}_n) + \lambda(\vec{w}_1 + \vec{w}_2 + \cdots + \vec{w}_n) \quad (\text{Distributive law of real numbers}) \\ &= \lambda\vec{v} + \lambda\vec{w}\end{aligned}$$

3.

$$\begin{aligned}\lambda(\vec{v} \cdot \vec{w}) &= \lambda(v_1 w_1 + v_2 w_2 + \cdots + v_n w_n) \\ &= \lambda(v_1 w_1) + \lambda(v_2 w_2) + \cdots + \lambda(v_n w_n) \quad (\text{Distributive law of real numbers}) \\ &= v_1(\lambda w_1) + v_2(\lambda w_2) + \cdots + v_n(\lambda w_n) \\ &= \vec{v} \cdot (\lambda\vec{w})\end{aligned}$$

4.

$$\begin{aligned}\vec{v} \cdot \vec{v} &= (v_1)^2 + (v_2)^2 + \cdots + (v_n)^2 \\ &= (\sqrt{(v_1)^2 + (v_2)^2 + \cdots + (v_n)^2})^2 \\ &= (\|\vec{v}\|)^2\end{aligned}$$

5. to be completed